

Redrob AI Candidate Ranker

Hackathon: Intelligent Candidate Discovery & Ranking Challenge

Team: Solo Submission

The Problem

Keyword matching doesn't work. The JD told us so.

- The JD explicitly states candidates are designed as traps -- profiles that fool keyword matchers.
- Keyword Stuffers: A Marketing Manager with 'Pinecone' and 'FAISS' in skills looks great to a keyword matcher but has zero production ML experience.
- Behavioral Ghosts: Candidates who list every skill but haven't logged in for months, have 120-day notice periods, and never respond to recruiters.
- Hidden Gems: Plain-language engineers who describe building ranking systems in career descriptions but don't keyword-stuff their skills section -- undervalued by keyword systems.

Our Insight

A recruiter doesn't score -- they reason.

Two questions a recruiter actually asks:

- (1) Did this person actually build this, or just list it?
- (2) Are they reachable right now?

We built a system that answers both. Not a scoring formula -- a reasoning engine that reads careers, checks evidence, and writes a narrative for each candidate.

System Architecture

Two phases: reason offline, rank fast.

OFFLINE (Pre-computation)

- JD decomposition into 8 requirements
- Feature extraction blueprints
- Career-substance keyword sets
- Honeypot impossibility rules

RANKING (Runtime)

- Pure Python, no API calls
- 43.7s on 100K candidates
- No network, no GPU
- Streaming JSONL, ~50MB RAM

Key constraint: No API calls. Pre-computation handles all reasoning.

The 6 Scoring Dimensions

Dimension	Weight	What It Measures
Career Substance	40%	Job titles + career descriptions vs JD requirements
Skill Credibility	22%	Endorsements + duration + corroboration
Behavioral Availability	15%	Recency + response rate + notice period
Experience Quality	11%	YoE band + stability + product companies
Star Predictor	7%	Title + company + ownership progression
Location	5%	Pune/Noida preferred per JD

The 4 Traps We Detect

Trap	Detection Method	Effect on Score
Keyword Stuffer	career_substance < 0.08	Hard-capped at 0.15
Honeypot	5 impossibility checks	Score = 0.0
Behavioral Ghost	behavioral_availability signals	Heavy down-weight
Hidden Gem	Career >> skills gap	+0.05 bonus

Shadow Recruiter

Every candidate gets a narrative, not a score dump.

Rank 1

Product-only career across Netflix, Meta, Mad Street Den (8yr). That's rare in this pool. The Elasticsearch signal is strong: expert proficiency, 44 endorsements, 96mo duration. Consistent with han...

Rank 6

Senior Machine Learning Engineer at Zomato, 7 years total. Closest JD-match role: Senior Machine Learning Engineer at Zomato (2yr 2mo). 50 people endorsed Learning to Rank (expert) across a 70-mont...

Rank 60

6 years entirely at product companies (Microsoft, Amazon) -- no services stint. Ranked 60 with composite 0.898. The skill credibility dimension (35%) is the gap -- career trajectory at 50% doesn't co...

Cohort Comparator

What the pool tells the hiring manager.

Recommendation: Pool has 2 unmet JD requirements and 1 shared gaps. Review the detailed sections before finalizing
100 candidates analyzed. Overall pool quality: ADEQUATE.
Outreach:

- 2 JD requirements unmet in top-10; 1 shared gap(s) across pool.
- Key gap: Early-stage / founding team experience -- 10/10 top candidates missing. 10/10 top candidates don't clearly satisfy this. This is a pool-wide supply gap, not individual weakness.
- Geographic spread looks healthy: 6 different cities in top-10.

Prove It Engine + Interview Blueprint

Rank 1 Candidate Deep Dive

Prove It Engine

[UNVERIFIED]

Advanced-level Speech Recognition

'Speech Recognition' not found in career descriptions; on...

[CORROBORATED]

Advanced-level PEFT

55 endorsements and 43 months of experience with 'PEFT'

[CORROBORATED]

Ownership claim: '...and ranking. most recently...

Title 'senior ai engineer' in career_history[0] suggests ...

[CORROBORATED]

States 7.8 years of experience

Career timeline totals 7.7 years, within 1.5yr of stated ...

Interview Questions

Q1:

Walk me through how you handled relevance tuning and query performance in Elasticsearch...

Why: The candidate's strongest credibility signal is Elasticsearch (endo...

Q2:

Our codebase is Python-heavy with strict performance requirements. Describe a time you ...

Why: Python is not prominently listed in the candidate's skill set. We n...

Q3:

Imagine it is your first day here. You inherit a ranking service that is live but under...

Why: No strong career-pattern signal was detected, so we use a general o...

Performance

Metric	Value
Runtime	43.7s / 300s limit
Memory	Streaming JSONL, ~50MB RAM
Honeypots Detected	10,001
Hidden Gems in Top-20	18
Validation	PASSED

Live Demo

`streamlit run demo/app.py`

All 6 panels load with real data:

- Shadow Recruiter -- Narrative reasoning for every candidate
- Prove It Engine -- Cross-reference claims vs evidence
- Deception Detector -- Honeypot & keyword stuffer flags
- Interview Blueprint -- Candidate-specific questions
- HM Translator -- Cohort analysis for hiring managers
- Signals Dashboard -- Behavioral availability metrics

What Makes This Different

- 1 We read careers, not keywords.
- 2 We reason, not score.
- 3 We help the recruiter, not replace them.